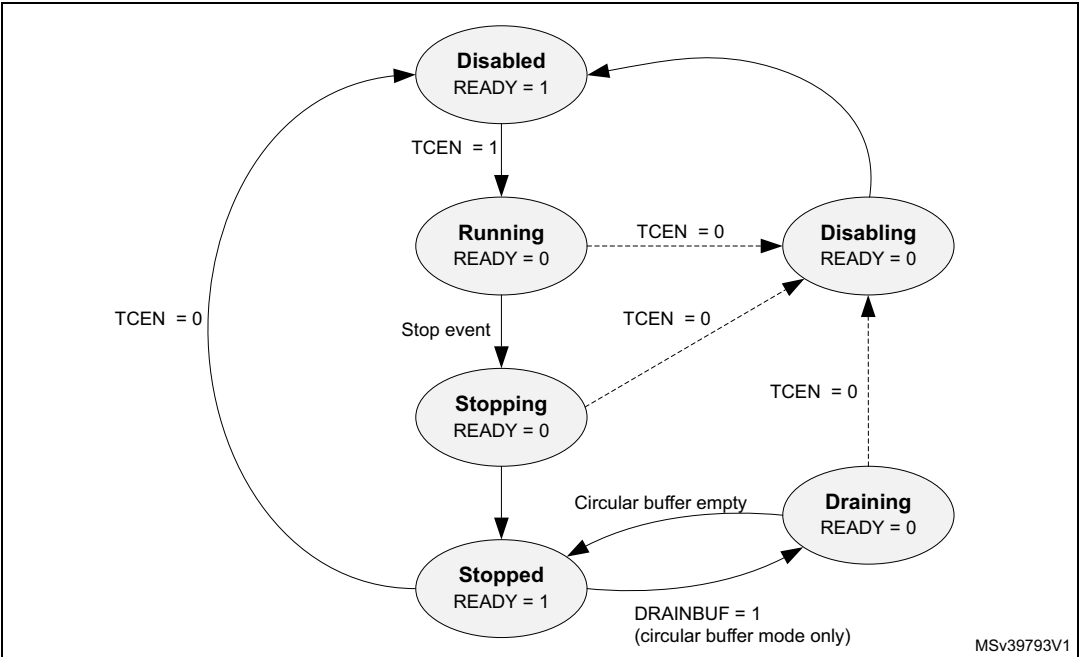


Figure 837. ETF state transition diagram



For more information on the CoreSight™ trace memory controller component, refer to the Arm® CoreSight™ trace memory controller technical reference manual [3].

ETF registers

ETF RAM size register (ETF_RSZ)

Address offset: 0x004

Reset value: 0x0000 0400

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	RSZ[30:16]														
	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSZ[15:0]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bit 31 Reserved, must be kept at reset value.

Bits 30:0 **RSZ[30:0]**: RAM size
 The value of the field indicates the number of 32-bit words
 0x400: 1024 words = 4 Kbytes

ETF status register (ETF_STS)

Address offset: 0x00C

Reset value: 0x0000 001C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	EMPTY	FEMPTY	READY	TRIGD	FULL
											r	r	r	r	r

Bits 31:5 Reserved, must be kept at reset value.

Bit 4 **EMPTY**: Trace FIFO empty

This bit is valid only when the TCEN bit of the ETF_CTL register is high. This bit reads as zero when TCEN is low.

0: Trace FIFO contains data

1: Trace FIFO is empty.

Note: Empty trace FIFO does not mean that the ETF pipeline is empty. The latter is indicated by the FEMPTY bit.

Bit 3 **FEMPTY**: Formatter empty

This bit is set when trace capture has stopped, and all internal pipelines and buffers have been drained. Unlike READY, it is not affected by buffer drains. The ACQCOMP output reflects the value of this bit.

Bit 2 **READY**: ETF ready

This bit is set when trace capture has stopped and all internal pipelines and buffers have been drained (Stopped or Disabled state)

Bit 1 **TRIGD**: Triggered

The Triggered bit is set when trace capture is in progress and the TMC has detected a Trigger Event. This bit is cleared when leaving Disabled state.

This bit is operational only in the Circular buffer mode. In all other modes, this bit is always low.

This bit does not indicate that a trigger has been embedded in the formatted output trace data from the TMC. Trigger indication on the output trace stream is determined by the programming of the Formatter and Flush Control Register, ETF_FFCR.

Bit 0 **FULL**: Trace buffer full

In circular buffer mode, this flag is set when the RAM write pointer wraps around the top of the buffer, and remains set until the TCEN bit of the ETF_CTL register is cleared and set.

In software and hardware FIFO modes, this flag indicates that the current space in the trace memory is less than or equal to the value programmed in the ETF_BUFWM Register, that is, Fill level $\geq$ MEM_SIZE - BUFWM.

This bit is cleared when leaving Disabled state. The FULL output reflects the value of this register bit.

ETF RAM read data register (ETF_RRD)

Address offset: 0x010

Reset value: Unknown

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RRD[31:16]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RRD[15:0]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:0 **RRD[31:0]**: RAM Read Data.

Circular buffer mode:

When in Stopped state and the buffer is not empty, reading this register returns the next word of data from the trace buffer. When the whole trace buffer has been read, the Empty bit in the ETF_STS Register is set, and subsequent reads return 0xFFFFFFFF. Reading this register when not in Stopped state returns 0xFFFFFFFF.

Software FIFO mode:

Reading this register returns data from the FIFO. If this register is read when the FIFO is empty, the data returned is 0xFFFFFFFF.

Hardware FIFO mode:

Reading this register returns 0xFFFFFFFF.

ETF RAM read pointer register (ETF_RRP)

Address offset: 0x014

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	RRP[12:0]												
			rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW

Bits 31:13 Reserved, must be kept at reset value.

Bits 12:0 **RRP[12:0]**: RAM Read Pointer

The RAM Read Pointer Register contains the value of the read pointer that is used to read entries from the trace memory over the APB interface via the ETF_RRD register. The pointer can be programmed with a byte address, 64-bit aligned (that is, bits 0 to 3 should be zero). The pointer is incremented by 8 each time a full 64-bit FIFO entry has been written. When the pointer reaches its maximum value, it wraps around.

This register can only be written to while in Disabled state. It can be read in Disabled state, in Stopped state in circular buffer mode and SW FIFO mode, and also in Running and Stopping states in SW FIFO mode.

ETF RAM write pointer register (ETF_RWP)

Address offset: 0x018

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	RWP[12:0]												
			rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW

Bits 31:13 Reserved, must be kept at reset value.

Bits 12:0 **RWP[12:0]**: RAM write pointer

The RAM write pointer register contains the value of the write pointer that is used to write entries into the trace memory over the APB interface via the ETF_RWD register. The pointer can be programmed with a byte address, 64-bit aligned (that is, bits 0 to 3 should be zero). The pointer is incremented by 8 each time a full 64-bit FIFO entry has been read. When the pointer reaches its maximum value, it wraps around.

This register can only be written to while in Disabled state. It can be read in Disabled state, in Stopped state in circular buffer mode and SW FIFO mode, and also in Running and Stopping states in SW FIFO mode.

ETF trigger counter register (ETF_TRG)

Address offset: 0x01C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	TRG[10:0]										
					rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW

Bits 31:11 Reserved, must be kept at reset value.

Bits 10:0 **TRG[10:0]**: Trigger counter

In Circular buffer mode, specifies the number of 32-bit words to capture in the trace RAM following the detection of either a rising edge on the TRIGIN input or a trigger packet in the incoming trace stream, ATID = 7'h7D. On capturing the specified number of data words, a trigger event occurs. The effect of a trigger event on the ETF behavior is controlled by the FFCR Register.

The number of 32-bit words written into the trace RAM following the trigger is the value stored in this register, plus one. This register is ignored when the ETF is in Software FIFO mode or Hardware FIFO mode. When the trigger counter starts counting, any additional triggers, either on TRIGIN or in the incoming trace stream, are ignored until the counter reaches zero. When the trigger counter has reached zero, it remains at zero until it is re-programmed with a write to this register.

This register is cleared when READY goes high, so that the state of the counter when trace capture has stopped does not affect a subsequent trace capture session. Writing to this register when not in Disabled state results in unpredictable behavior.

A read access to this register is permitted at any time when in Disabled state, or in Circular buffer mode. A read access returns the current value of the trigger counter.

ETF control register (ETF_CTL)

Address offset: 0x020

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TCEN
															rw

Bits 31:1 Reserved, must be kept at reset value.

Bit 0 **TCEN**: Trace capture enable

When writing:

0: Disable trace capture (moves from Running, Stopping or Stopped state into Disabling or Disabled state)

1: Enable trace capture (moves from Disabled state to Running state)

When reading, this bit is low when in Disabling or Disabled states, and high otherwise.

ETF RAM write data register (ETF_RWD)

Address offset: 0x024

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RWD[31:16]															
w	w	w	w	w	w	w	w	w	w	w	w	w	w	w	w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RWD[15:0]															
w	w	w	w	w	w	w	w	w	w	w	w	w	w	w	w

Bits 31:0 **RWD[31:0]**: RAM write data

When in Disabled state, a write to this register stores data at the location pointed to by the RWP. Writes to this register when not in Disabled state are ignored. When a full memory width (64-bit) of data has been written, the data is written to memory and the RAM Write Pointer is incremented to the next memory word.

This register is used for test purposes.

ETF mode register (ETF_MODE)

Address offset: 0x028

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MODE[1:0]	
														rw	rw

Bits 31:2 Reserved, must be kept at reset value.

Bits 1:0 **MODE[1:0]**: Operation mode

00b: Circular buffer mode

In this mode, the trace memory is used as a circular buffer. Trace data is captured into the Trace memory starting from the location pointed to by the write pointer register. Even when the trace memory is full, incoming trace data continues to be overwritten into the trace memory until a stop condition has occurred.

01b: Software FIFO mode

In this mode, the trace memory is used as a FIFO that can be read through the RRD Register while trace is being captured. Trace data is captured into the trace RAM and when full, the incoming trace stream is stalled.

10b: Hardware FIFO mode

In this mode, the trace memory is used as a FIFO that is drained through the ATB master interface. Trace data is captured into the trace RAM and when full, the incoming trace stream is stalled. When the trace buffer is non-empty, trace data is drained out through the ATB master interface to the TPIU.

11b: Reserved

ETF latched buffer fill level register (ETF_LBUFLVL)

Address offset: 0x02C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	LBUFLEVEL[11:0]											
				r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:12 Reserved, must be kept at reset value.

Bits 11:0 **LBUFLEVEL[11:0]**: Latched buffer fill level

Reading this register returns the maximum fill level of the trace memory in 32-bit words since this register was last read. Reading this register also results in its contents being updated to the current fill level.

When entering Disabled state, this register retains its last value. While in Disabled state, reads from this register do not affect its value. When exiting Disabled state, the LBUFLEVEL register is cleared.

This register is used for performance analysis of the trace system.

ETF current buffer fill level register (ETF_CBUFLVL)

Address offset: 0x030

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	CBUFLEVEL[11:0]											
				r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:12 Reserved, must be kept at reset value.

Bits 11:0 **CBUFLEVEL[11:0]**: Current buffer fill level

Reading this register returns the current fill level of the trace memory in 32-bit words.

This register is cleared when TCEN is low.

ETF buffer level watermark register (ETF_BUFWM)

Address offset: 0x034

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	BUFWM[10:0]										
					r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bits 31:11 Reserved, must be kept at reset value.

Bits 10:0 **BUFWM[10:0]**: Buffer level watermark

The value programmed into this register indicates the required threshold vacancy level in the trace memory in 32-bit words. When the space in the FIFO is less than or equal to this value, that is, Fill level $\geq$ MEM_SIZE - BUFWM, the FULL output is pulled high and the FULL bit in the STS Register is set.

This register is used only in Software FIFO and Hardware FIFO modes. In Circular buffer mode, this functionality can be obtained by programming the RWP to the required vacancy trigger level, so that when the pointer wraps around, the FULL bit is set indicating that the vacancy level has fallen below the required level.

The maximum value that can be written into this register is MEM_SIZE - 1. In this case, the FULL bit output is asserted after the first 32-bit word is written to trace memory.

Writing to this register other than when in disabled state results in unpredictable behavior.

ETF formatter and flush status register (ETF_FFSR)

Address offset: 0x300

Reset value: 0x0000 0002

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	FTSTO PPED	FLINPR OG
														r	r

Bits 31:2 Reserved, must be kept at reset value.

Bit 1 **FTSTOPPED**: Formatter stopped

This bit behaves in the same way as the FEMPTY bit in the ETF_STS register.

Bit 0 **FLINPROG**: Flush in progress

Indicates whether a flush on the ATB slave port is in progress. This bit reflects the status of the AFVALIDS output. A flush can be initiated by the flush control bits in the ETF_FFCR register, or requested by the ATB master port.

0: No flush in progress

1: Flush in progress

ETF formatter and flush control register (ETF_FFCR)

Address offset: 0x304

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	DRAIN BUF.	STPO NTRG EV.	STOP ONFL	Res.	TRIGO NFL.	TRGON TRGEV	TRGON TRGIN	Res.	FLUSH MAN	FONTR GEV.	FONFLI N	Res.	Res.	ENTI	ENFT
	rw	rw	rw		rw	rw	rw		rw	rw	rw			rw	rw

Bits 31:15 Reserved, must be kept at reset value.

Bit 14 **DRAINBUF**: Drain buffer

This bit is used to enable draining of the trace data through the ATB master interface after the formatter has stopped. This is useful in Circular buffer mode to capture trace data into trace memory and then to drain the captured trace through the ATB master interface.

Writing a one to this bit when in Stopped state starts the drain of the trace buffer content through the ATB Master interface. This bit always reads as zero. The READY bit in the ETF_STS register goes low while the drain is in progress.

This bit is only functional when the ETF is in Circular buffer mode and formatting is enabled, that is, the ENFT bit in the ETF_FFCR register is set. **Setting this bit when the ETF is in any other mode, or when not in Stopped state, results in Unpredictable behavior.**

When trace capture is complete in Circular buffer mode, all of the captured trace must be retrieved from the trace memory through the same mechanism, either read all trace data out through RRD reads, or drain all trace data by setting the DRAINBUF bit. Setting the DRAINBUF bit after some of the captured trace has been read out through RRD results in unpredictable behavior.

Bit 13 **STPONTRGEV**: Stop on trigger event

0: No effect

1: Stop trace capture when a trigger event occurs

Enabling the ETF in Software FIFO mode or Hardware FIFO mode with this bit set results in unpredictable behavior.

Bit 12 **STOPONFL**: Stop on flush

0: No effect

1: Stop trace capture when flush is completed

If a flush is initiated by the ATB master interface, its completion does not lead to a formatter stop regardless of the value programmed in this bit.

Bit 11 Reserved, must be kept at reset value.

Bit 10 **TRIGONFL**: Trigger on flush

0: No effect

1: Indicate a trigger in the trace stream when flush is completed

If ENFT and ENTI are both clear, this bit is ignored and no trigger is inserted into the trace stream.

If a flush is initiated by the ATB master interface, its completion does not lead to a trigger indication regardless of the value programmed in this bit.

Bit 9 **TRGONTRGEV**: Trigger on trigger event

0: No effect

1: Indicate a trigger in the trace stream when trigger event occurs

If ENFT and ENTI are both clear, this bit is ignored and no trigger is inserted into the trace stream.

This bit is not supported in Software FIFO mode or Hardware FIFO mode.

Bit 8 **TRGONTRGIN**: Trigger on trigger in

0: No effect

1: Indicate a trigger in the trace stream when a rising edge is detected on the TRIGIN input.

If ENFT and ENTI are both clear, this bit is ignored and no trigger is inserted into the trace stream.

Bit 7 Reserved, must be kept at reset value.

Bit 6 **FLUSHMAN**: Manual flush

0: No effect

1: Flush the trace FIFO and pipeline

This bit is cleared automatically when the flush completes. If the TCEN bit in the ETF_CTL register is 0, writes to this bit are ignored.

Bit 5 **FONTRGEV**: Flush on trigger event

0: No effect

1: Flush the trace FIFO and pipeline if a trigger event occurs

This bit is not supported in Software FIFO mode or Hardware FIFO mode. If STPONTRGEV is set, this bit is ignored.

Bit 4 **FONFLIN**: Flush on flush in

0: No effect

1: Flush the trace FIFO and pipeline if when a rising edge is detected on the FLUSHIN input

Bits 3:2 Reserved, must be kept at reset value.

Bit 1 **ENTI**: Enable trigger insertion

Setting this bit enables the insertion of triggers in the formatted trace stream. A trigger is flagged by inserting one byte of data 8'h00 with ATID 7'h7D in the trace stream. Trigger indication on the trace stream is further controlled by the register bits TRIGONFL, TRGONTRGEV, and TRGONTRGIN in the FFCR Register. **This bit can only be changed when READY is high, and TCEN is low. This bit takes effect only when the ENFT register bit in this register is set. If ENTI bit is set to high when ENFT is low, it results in formatting being enabled.**

Bit 0 **ENFT**: Enable formatting.

0: Formatting is disabled. Incoming trace data is assumed to be from a single trace source.
1: Formatting is enabled.

If multiple ATIDs are received by the ETF when trace capture is enabled and the formatter is disabled, an interleaving of trace data occurs. Disabling of formatting is supported only in Circular buffer mode. **If the ETF is enabled in a mode other than Circular buffer mode with ENFT low, then formatting is enabled. If ENTI bit is set to high when ENFT is low, it results in formatting being enabled.**

This bit is ignored when in Disabled state.

ETF periodic synchronization counter register (ETF_PSCR)

Address offset: 0x308

Reset value: 0x0000 000A

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PSCOUNT[4:0]				
											rw	rw	rw	rw	rw

Bits 31:5 Reserved, must be kept at reset value.

Bits 4:0 **PSCOUNT[4:0]**: Synchronization counter reload value

Determines the reload value of the Synchronization Counter. The reload value takes effect the next time the counter reaches zero. Reads from this register return the reload value programmed into this register. This register is set to 0xA on reset, corresponding to a synchronization period of 1024 bytes.

0x0: Synchronization disabled

0x1-0x6: Reserved

0x7-0x1B: Synchronization period is 2^{PSCOUNT} bytes

0x1C-0x1F: Reserved

Reset value: 0x0000 000F

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CLAIMSET[3:0]			
												rw	rw	rw	rw

0xF: Indicates there are four bits in claim tag

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CLAIMCLR[3:0]			
												rw	rw	rw	rw

Read: Returns current value of claim tag

ETF lock access register (ETF_LAR)

Address offset: 0xFB0

Reset value: N/A

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ACCESS_W[31:16]															
w	w	w	w	w	w	w	w	w	w	w	w	w	w	w	w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ACCESS_W[15:0]															
w	w	w	w	w	w	w	w	w	w	w	w	w	w	w	w

Bits 31:0 **ACCESS_W[31:0]**: ETF register access enable

Enables write access to some ETF registers by processor cores (debuggers do not need to unlock the component)

0xC5ACCE55: Enable write access

Other values: Disable write access

ETF lock status register (ETF_LSR)

Address offset: 0xFB4

Reset value: 0x0000 0003

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	LOCK TYPE	LOCK GRANT	LOCK EXIST
													r	r	r

Bits 31:3 Reserved, must be kept at reset value.

Bit 2 **LOCKTYPE**: Size of the ETF_LAR register

0: 32-bit

Bit 1 **LOCKGRANT**: Current status of lock

This bit always returns zero when read by an external debugger.

0: Write access is permitted

1: Write access is blocked. Only read access is permitted.

Bit 0 **LOCKEXIST**: Existence of lock control mechanism

The bit indicates whether a lock control mechanism exists. It always returns zero when read by an external debugger.

0: No lock control mechanism exists

1: Lock control mechanism is implemented

ETF authentication status register (ETF_AUTHSTAT)

Address offset: 0xFB8

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SNID[1:0]		SID[1:0]		NSNID[1:0]		NSID[1:0]	
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:6 **SNID[1:0]**: Security level for secure non-invasive debug

0x0: Not implemented

Bits 5:4 **SID[1:0]**: Security level for secure invasive debug

0x0: Not implemented

Bits 3:2 **NSNID[1:0]**: Security level for non-secure non-invasive debug

0x0: Not implemented

Bits 1:0 **NSID[1:0]**: Security level for non-secure invasive debug

0x0: Not implemented

ETF device configuration register (ETF_DEVID)

Address offset: 0xFC8

Reset value: 0x0000 01C0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	MEMWIDTH[2:0]			CONFIGTYP[1:0]		CLK SCHEM	ATBINPORTCNT[4:0]				
					r	r	r	r	r	r	r	r	r	r	r

Bits 31:11 Reserved, must be kept at reset value.

Bits 10:8 **MEMWIDTH[2:0]**: Memory interface data bus width

0x3: 64 bits (corresponds to 32-bit ATB data)

Bits 7:6 **CONFIGTYP[1:0]**: Configuration type of component (ETB, ETR or ETF)

0x2: ETF

Bit 5 **CLKSCHEM**: RAM clocking scheme (synchronous or asynchronous)

0: Synchronous

Bits 4:0 **ATBINPORTCNT[4:0]**: Number/type of ATB input port multiplexing

0x0: None

ETF device type identifier register (ETF_DEVTYPE)

Address offset: 0xFCC

Reset value: 0x0000 0032

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SUBTYPE[3:0]				MAJORTYPE[3:0]			
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:4 **SUBTYPE[3:0]**: Sub-classification

0x3: Captures trace data from the ATB slave interface into RAM that can be drained through the ATB master interface

Bits 3:0 **MAJORTYPE[3:0]**: Major classification

0x2: Component is a trace link because it has an ATB master interface through which trace data can be drained out in Hardware FIFO mode.

ETF CoreSight peripheral identity register 4 (ETF_PIDR4)

Address offset: 0xFD0

Reset value: 0x0000 0004

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	4KCOUNT[3:0]				JEP106CON[3:0]			
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:4 **4KCOUNT[3:0]**: Register file size

0x0: Register file occupies a single 4 Kbyte region

Bits 3:0 **JEP106CON[3:0]**: JEP106 continuation code

0x4: Arm® JEDEC code

ETF CoreSight peripheral identity register 0 (ETF_PIDR0)

Address offset: 0xFE0

Reset value: 0x0000 0061

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PARTNUM[7:0]							
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:0 **PARTNUM[7:0]**: Part number field, bits [7:0]

0x61: ETF part number

ETF CoreSight peripheral identity register 1 (ETF_PIDR1)

Address offset: 0xFE4

Reset value: 0x0000 00B9

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	JEP106ID[3:0]				PARTNUM[11:8]			
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:4 **JEP106ID[3:0]**: JEP106 identity code field, bits [3:0]

0xB: Arm® JEDEC code

Bits 3:0 **PARTNUM[11:8]**: Part number field, bits [11:8]

0x9: ETF part number

ETF CoreSight peripheral identity register 2 (ETF_PIDR2)

Address offset: 0xFE8

Reset value: 0x0000 001B

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	REVISION[3:0]				JEDEC	JEP106ID[6:4]		
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:4 **REVISION[3:0]**: Component revision number
0x1: r0p1

Bit 3 **JEDEC**: JEDEC assigned value
1: Designer ID specified by JEDEC

Bits 2:0 **JEP106ID[6:4]**: JEP106 identity code field, bits [6:4]
0x3: Arm® JEDEC code

ETF CoreSight peripheral identity register 3 (ETF_PIDR3)

Address offset: 0xFEC

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	REVAND[3:0]				CMOD[3:0]			
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:4 **REVAND[3:0]**: Metal fix version
0x0: No metal fix

Bits 3:0 **CMOD[3:0]**: Customer modified
0x0: No customer modifications

ETF CoreSight component identity register 0 (ETF_CIDR0)

Address offset: 0xFF0

Reset value: 0x0000 000D

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PREAMBLE[7:0]							
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:0 **PREAMBLE[7:0]**: Component ID field, bits [7:0]
0x0D: Common ID value

ETF CoreSight component identity register 1 (ETF_CIDR1)

Address offset: 0xFF4

Reset value: 0x0000 0090

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CLASS[3:0]				PREAMBLE[11:8]			
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:4 **CLASS[3:0]**: Component ID field, bits [15:12] - component class

0x9: CoreSight component

Bits 3:0 **PREAMBLE[11:8]**: Component ID field, bits [11:8]

0x0: Common ID value

ETF CoreSight component identity register 2 (ETF_CIDR2)

Address offset: 0xFF8

Reset value: 0x0000 0005

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PREAMBLE[19:12]							
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:0 **PREAMBLE[19:12]**: Component ID field, bits [23:16]

0x05: Common ID value

ETF CoreSight component identity register 3 (ETF_CIDR3)

Address offset: 0xFFC

Reset value: 0x0000 00B1

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PREAMBLE[27:20]							
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:0 **PREAMBLE[27:20]**: Component ID field, bits [31:24]

0xB1: Common ID value

ETF register map and reset values

Table 615. ETF register map and reset values

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0				
0x004	ETF_RSZ	Res	RSZ[30:0]																																		
	Reset value		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0				
0x008	Reserved	Reserved																																			
0x00C	ETF_STS	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	EMPTY	FEMPTY	READY	TRIGD	FULL				
	Reset value																												1	1	1	0	0				
0x010	ETF_RRD	RRD[31:0]																																			
	Reset value	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-					
0x014	ETF_RRP	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	RRP[12:0]																
	Reset value																				0	0	0	0	0	0	0	0	0	0	0	0	0				
0x018	ETF_RWP	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	RWP[12:0]																
	Reset value																				0	0	0	0	0	0	0	0	0	0	0	0	0				
0x01C	ETF_TRG	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	TRG[10:0]															
	Reset value																							0	0	0	0	0	0	0	0	0	0	0			
0x020	ETF_CTL	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	TCEN				
	Reset value																																0				
0x024	ETF_RWD	RWD[31:0]																																			
	Reset value	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-				
0x028	ETF_MODE	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	MODE[1:0]					
	Reset value																															0	0				
0x02C	ETF_LBUFLVL	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	LBUFLVL[11:0]															
	Reset value																				0	0	0	0	0	0	0	0	0	0	0	0	0				
0x030	ETF_CBUFLVL	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	CBUFLVL[11:0]															
	Reset value																				0	0	0	0	0	0	0	0	0	0	0	0	0				
0x034	ETF_BUFWM	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	BUFWM[10:0]															
	Reset value																					0	0	0	0	0	0	0	0	0	0	0	0				
0x038 to 0x2FC	Reserved	Reserved																																			
0x300	ETF_FFSR	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	FTSTOPPED	FLINPROG				
	Reset value																															1	0				

Table 615. ETF register map and reset values (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0				
0x304	ETF_FFCR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	DRINBUF	STPONTRGEV	STOPONFL	Res.	TRGONFL	TRGONTRGEV	TRGONTRGIN	Res.	FLUSHMAN	FONTRGEV	FONFLIN	Res.	Res.	ENTI	ENFT			
	Reset value																		0	0	0		0	0	0		0	0	0			0	0				
0x308	ETF_PSCR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PSCOUNT[4:0]							
	Reset value																												0	1	0	1	0				
0x30C to 0xF9C	Reserved	Reserved																																			
0xFA0	ETF_CLAIMSET	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CLAIMSET[3:0]						
	Reset value																													1	1	1	1				
0xFA4	ETF_CLAIMCLR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CLAIMCLR[3:0]						
	Reset value																													0	0	0	0				
0xFA8 to 0xFAC	Reserved	Reserved																																			
0xFB0	ETF_LAR	ACCESS_W[31:0]																																			
	Reset value	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-				
0xFB4	ETF_LSR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	LOCKTYPE	LOCKGRANT	LOCKEXIST			
	Reset value																														0	1	1				
0xFB8	ETF_AUTHSTAT	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	NSID[1:0]	NSID[1:0]	NSID[1:0]			
	Reset value																														0	0	0				
0xFBC to 0xFC4	Reserved	Reserved																																			
0xFC8	ETF_DEVID	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MEMWDTH[2:0]				CONFIGTYP[1:0]		CLKSCHEM		ATBINPORTCNT[4:0]					
	Reset value																						0	0	1	1	1	0	0	0	0	0	0				
0xFCC	ETF_DEVTYPE	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SUBTYPE[3:0]				MAJORTYPE[3:0]							
	Reset value																									0	0	1	1	0	0	1	0				

Table 615. ETF register map and reset values (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0xFD0	ETF_PIDR4	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	4KCOUNT[3:0]				JEP106CON[3:0]			
	Reset value																									0	0	0	0	0	1	0	0
0xFD4 to 0xFDC	Reserved	Reserved																															
0xFE0	ETF_PIDR0	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PARTNUM[7:0]							
	Reset value																									0	1	1	0	0	0	0	1
0xFE4	ETF_PIDR1	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	JEP106ID[3:0]				PARTNUM[11:8]			
	Reset value																									1	0	1	1	1	0	0	1
0xFE8	ETF_PIDR2	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	REVISION[3:0]				JEDEC		JEP106ID[6:4]	
	Reset value																									0	0	0	1	1	0	1	1
0xFEC	ETF_PIDR3	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	REVAND[3:0]				CMOD[3:0]			
	Reset value																									0	0	0	0	0	0	0	0
0xFF0	ETF_CIDR0	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PREAMBLE[7:0]							
	Reset value																									0	0	0	0	1	1	0	1
0xFF4	ETF_CIDR1	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CLASS[3:0]				PREAMBLE[11:8]			
	Reset value																									1	0	0	1	0	0	0	0
0xFF8	ETF_CIDR2	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PREAMBLE[19:12]							
	Reset value																									0	0	0	0	0	1	0	1
0xFFC	ETF_CIDR3	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PREAMBLE[27:20]							
	Reset value																									1	0	1	1	0	0	0	1

60.5.6 Trace port interface unit (TPIU)

The TPIU is a CoreSight™ component that formats the trace stream and outputs it on the external trace port signals. The TPIU has a single ATB slave port for incoming trace data. The trace port is a synchronous parallel port, comprising a clock output, TRACECK, and four data outputs, TRACED(7:0). The trace port width is programmable in the range 1 to 8. Using a smaller port width reduces the number of test points/connector pins needed, and frees up IOs for other purposes. However it restricts the bandwidth of the trace port and hence the quantity of trace information that can be output in real time. **The TRACECK output must be enabled by setting the TRACECLKEN bit in the DBGMCU_CR register before trace is sent to the TPIU. Furthermore, the TRACECK frequency can be programmed in the RCC.**

For more information on the Trace port interface CoreSight™ component, refer to the Arm® CoreSight™ SoC-400 technical reference manual [2].